

Chapter 4 - Affine reflection groups

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3 Alcoves

Exercise 3.1 [Trivial]

An alcove A consists of all $\lambda \in V$ satisfying strict inequalities $k_\alpha < (\lambda, \alpha) < k_\alpha + 1$, where α runs over Φ^+ and $k_\alpha \in \mathbf{Z}$. Its **upper closure** consists of those λ satisfying the inequalities obtained by replacing the second $<$ by $\leq$ in each case. Prove that each $\lambda \in V$ lies in the upper closure of a unique alcove.

Proof. (Check the definition of those walls.)

□